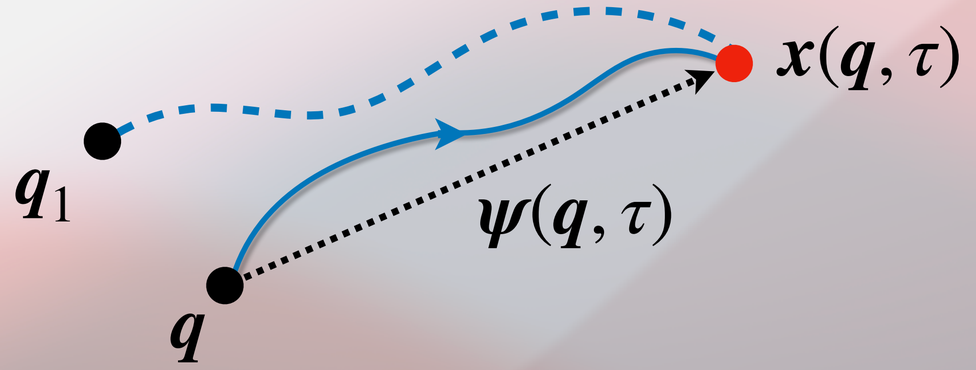
Cosmological shell-crossing using post-collapse perturbation theory

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KEK, Theory center, Journal club 2026.09.25



Based on:

[1] A. Taruya, S. Colombi, “Post-collapse perturbation theory in 1D cosmology - beyond shell-crossing” (2017)
[2] S. Saga, S. Colombi, A. Taruya, C. Rampf, A. Parichha, “Post-collapse Lagrangian perturbation theory in three dimensions” (2025)

Outline

Burgers equation

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0$$

Multi-valued velocity

Shell crossing

$$f(\boldsymbol{x}, \boldsymbol{v}) = \sum_s \rho_s(\boldsymbol{x})\delta_D[\boldsymbol{v} - \boldsymbol{v}_s(\boldsymbol{x})]$$

Single-stream PT breaks down at shell crossing

Post-collapse PT

$$x(q,\tau) = x^{\mathrm{BG}} + \delta x,$$
$$u(q,\tau) = u^{\mathrm{BG}} + \delta u$$

Post-collapse correction to LPT background

Introduction: inviscid Burgers equation

$u(x,t)$: fluid mean velocity

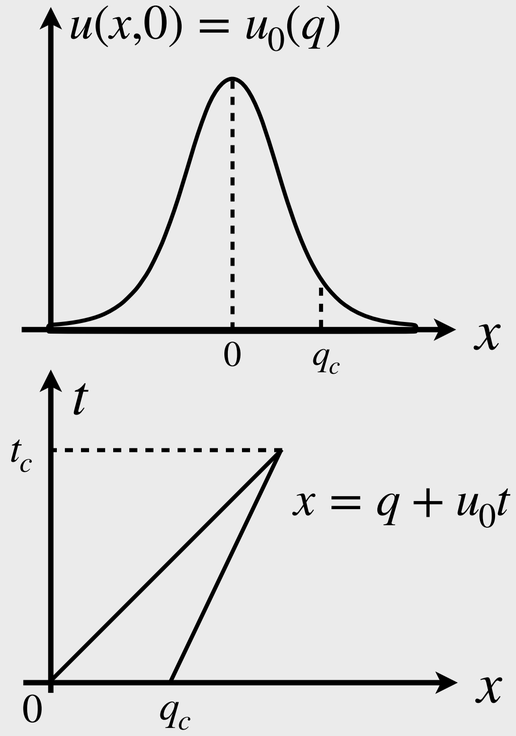
$u_0(x)$: initial condition

q: Lagrangian coordinates

- First-order nonlinear equation, $\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0$
- $u = u_0(x) = u(x,0) = const$, FourierT gives $u(x,t) = u_0(x - ut)$
- Using $\frac{du}{dt} = \frac{\partial u}{\partial t} + \frac{dx}{dt}\frac{\partial u}{\partial x} = 0$, along the curves $dx/dt = u$
- Characteristic curve: $q \equiv x - ut \rightarrow u(x,t) = u_0(q)$,
- Then, $J = \partial x/\partial q = 1 + t\,u_0'(q)$,

  as $J > 0$, single-valued;

  once $J = 0 \rightarrow t_c = -1/u_0'(q_c)$, multi-valued ($t > t_c$).



Inviscid Burger's equation (example)

$$u(x,0) = \operatorname{sech}^2(x)$$

$$\downarrow$$

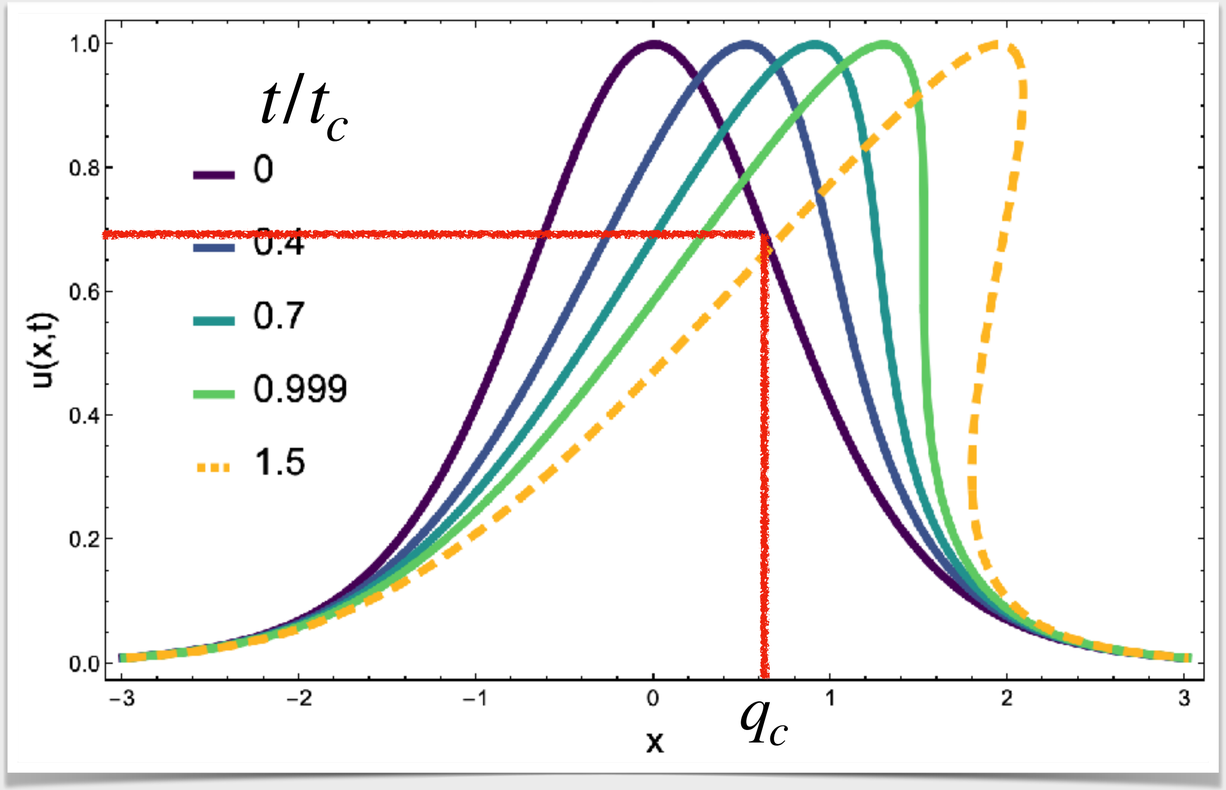
$$u'(x,0) = -2\operatorname{sech}^2(x)\tanh(x)$$

$$\downarrow \qquad \min J = 1 + t \ \min u_0'(q)$$

$$t_c = -1/u_0'(q_c) = \frac{3\sqrt{3}}{4}$$

$$q_c = \operatorname{arctanh}(1/\sqrt{3}) \approx 0.66$$

First crossing time & initial position



CDM shell-crossing

Before

$$f(\boldsymbol{x},\boldsymbol{v}) = \rho(\boldsymbol{x})\,\delta_D[\boldsymbol{v}-\boldsymbol{u}], \quad \sigma_{ij}=0$$

After

$$\sigma_{ij}^{(s)} = 0$$

$$f(\boldsymbol{x},\boldsymbol{v}) = \sum_s \rho_s(\boldsymbol{x})\,\delta_D[\boldsymbol{v}-\boldsymbol{v}_s], \quad \sigma_{ij}\neq 0$$

$f(\boldsymbol{x},\boldsymbol{v})$: phase space distribution

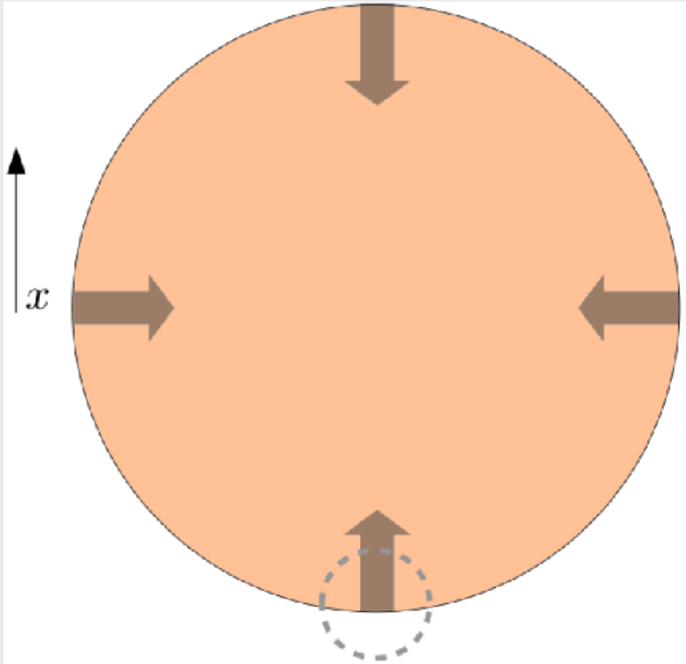
$\rho(\boldsymbol{x})$: fluid density

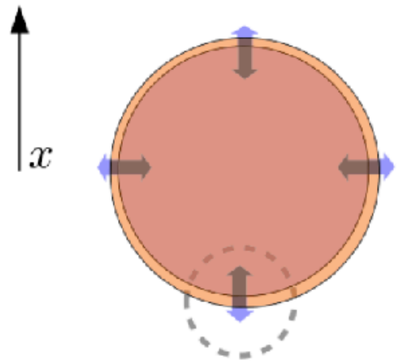
$\boldsymbol{v}_s$: single stream velocity

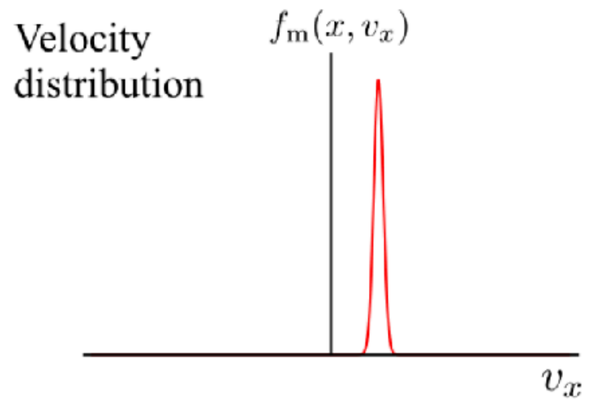
σ_{ij}: velocity dispersion

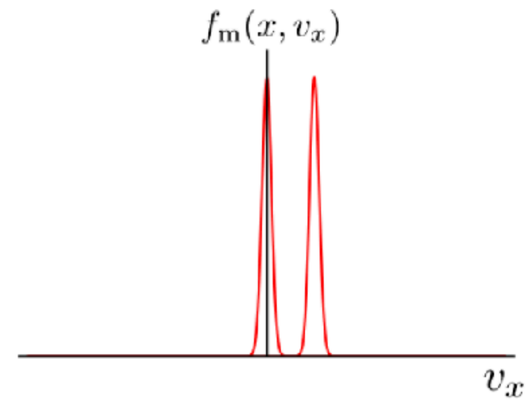
$$\sigma_{ij} = \frac{1}{\rho}\int d^3v\ (v_i-u_i)(v_j-u_j)\,f$$

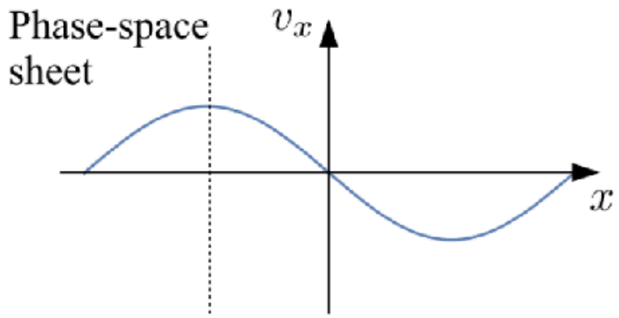
From "Modern Cosmology", figure 12.5

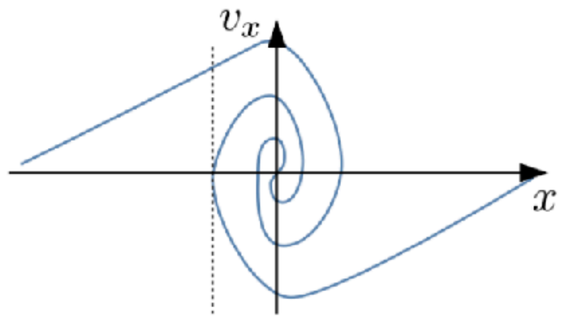












Multi-streaming (F=ma)

Fluid carrying itself around

Acceleration

Universe expansion

$$\frac{\partial u_i}{\partial t} + \frac{1}{a} u_j \partial_j u_i + H u_i = -\frac{1}{a}\partial_i \Phi - \frac{1}{a\rho}\partial_j(\rho \sigma_{ij})$$

Gravity

Velocity dispersion

$$u_t + u u_x = 0$$

Inviscid Burgers equation

Eulerian Vlasov moment equation

Eulerian perturbation theory

$\sigma_{ij} \neq 0$, effective field theory / stress tensor $\nabla \cdot \tau_{ij}^{\text{eff}}$, integrate out shell-crossing

Lagrangian perturbation theory

$q \mapsto x(q), v(q)$

Before crossing

Pre-collapse background flow

Zel'dovich / higher-order, single fluid

$\sigma_{ij} = 0$

After crossing

Post-collapse LPT

Multiple cold streams,

$\sigma_{ij}^{\text{total}} \neq 0, \sigma_{ij}^{(s)} = 0$

Pre-collapse (Lagrangian dynamics / Zel'dovich solution)

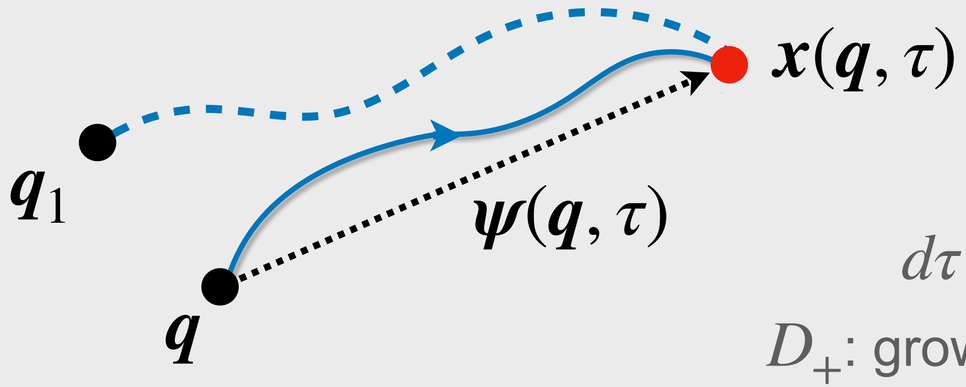
- Map to Lagrangian coordinates, $\boldsymbol{x}(\boldsymbol{q},\tau) = \boldsymbol{q} + \boldsymbol{\psi}(\boldsymbol{q},\tau) \rightarrow \boldsymbol{u} = \frac{d\boldsymbol{\psi}}{d\tau}$

exact solution for 1D

- Expand $\boldsymbol{\psi}(\boldsymbol{q},\tau) = \sum_{n=1}^{\infty} \boldsymbol{\psi}^{(n)}(\boldsymbol{q},\tau)$, factorize 1st $\boldsymbol{\psi}^{(1)}(\boldsymbol{q},\tau) = D_+(\tau)\, \boldsymbol{\psi}^{(1)}(\boldsymbol{q})$

- Linear solution: $\boldsymbol{x}(\boldsymbol{q},\tau) = \boldsymbol{q} + D_+(\tau)\boldsymbol{\psi}^{(1)}(\boldsymbol{q}) \rightarrow\ J_{ij}(\boldsymbol{q},\tau) = \delta_{ij} + D_+(\tau)\frac{\partial \psi_i^{(1)}(\boldsymbol{q})}{\partial q_j}$

- Before crossing, $\boldsymbol{q} \leftrightarrow \boldsymbol{x}$ (background flow)
  - first occur, $\boldsymbol{q} = \boldsymbol{q}_0$, $\det J(\boldsymbol{q}_0, \tau_0) = 0$
    - after crossing, $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3, \ldots \mapsto \boldsymbol{x}$



$d\tau \equiv a^{-2}dt$

D_+: growth factor

$\boldsymbol{\psi}$: displacement field

Then, near 1st crossing , add corrected background motion w/ multi-stream backreaction →

Post-collapse in 1D

q_0, τ_0: first-crossing Lagrangian label & super-conformal time

A, B, C: Taylor coefficients

- $J(q,\tau) = \dfrac{\partial x}{\partial q}$, first shell crossing $\rightarrow \min_q J = 0$:

$$J(q_0,\tau_0) = 0, \quad \partial_q J(q_0,\tau_0) = 0, \quad \partial_q^2 J(q_0,\tau_0) > 0$$

  - $x_q = 0, \quad x_{qq} = 0, \quad x_{qqq} > 0$

- Cubic approximation: $x^{\mathrm{BG}} - A \simeq -BQ + CQ^3$, $Q \equiv q - q_0$

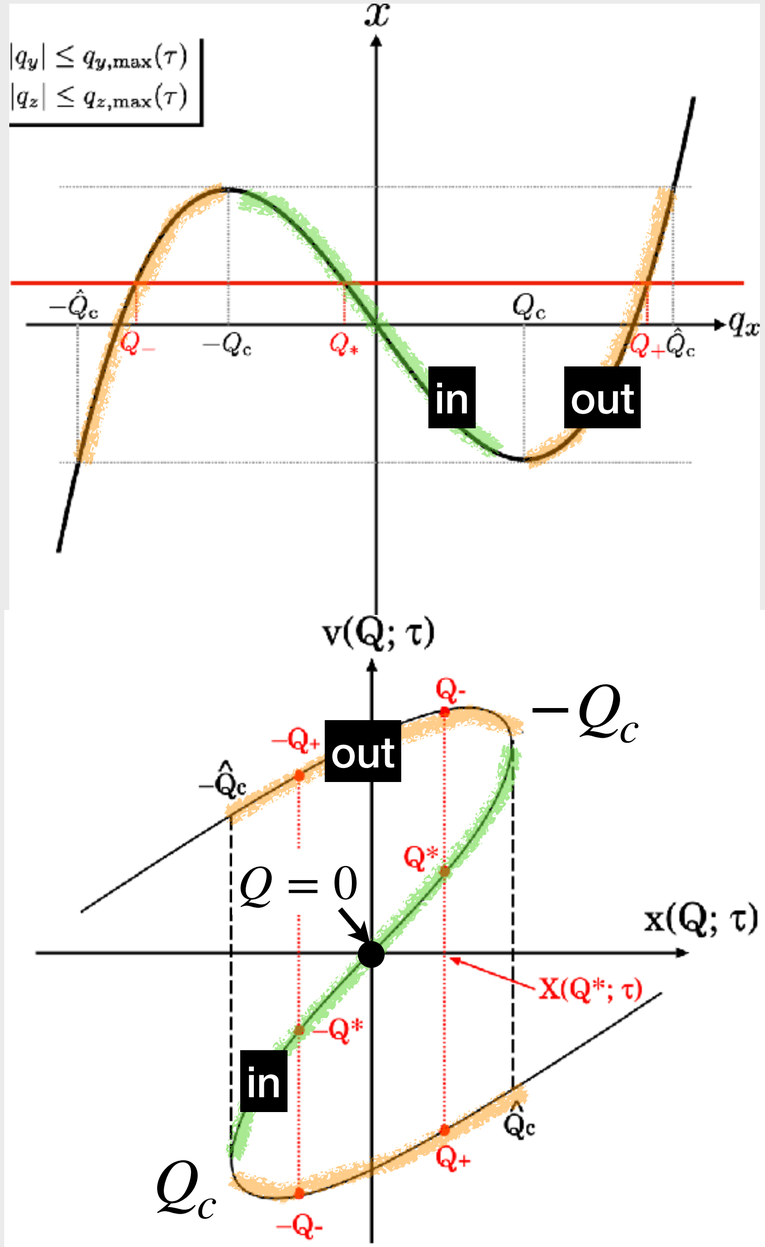
  - $J = -B + 3CQ^2 = 0 \rightarrow Q_c = \sqrt{B/3C}$ (caustic)

- $CQ^3 - BQ + A - x^{\mathrm{BG}} = 0 \rightarrow$ roots: $Q_- < Q_* < Q_+$

- Fixed Q ,

  $|Q| = \hat{Q}_c(\hat{\tau}_c)$ first enters multi-stream (out)

  $|Q| = Q_c(\tau_c)$ itself reaches caustic (in)



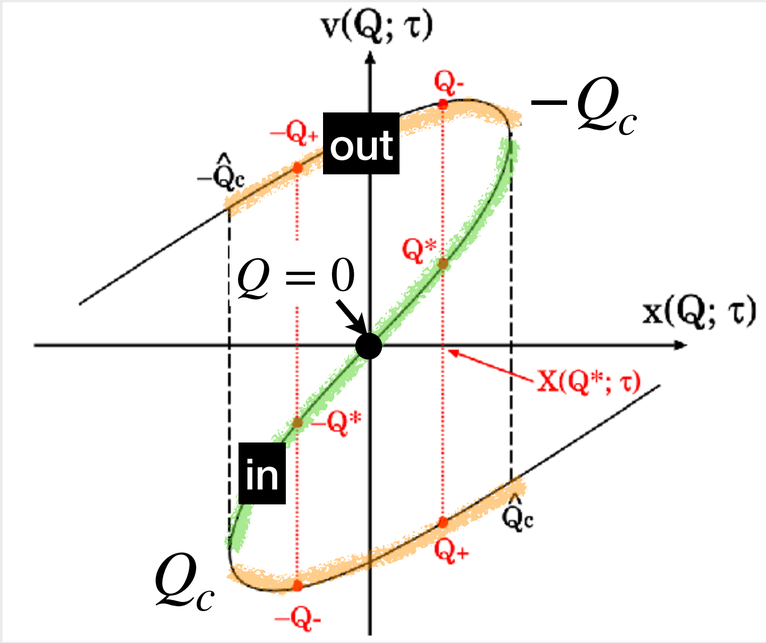
Correction to force, velocities, position

$\tau < \hat{\tau}_c(q)$ —— $x = x^{\rm BG}(q,\tau) = q + \psi(q) D_+(\tau)$ $\quad u = u^{\rm BG}(q,\tau) = \psi(q)\dfrac{D_+(\tau)}{d\tau}$

|- ballistic -|

$\hat{\tau}_c < \tau < \tau_c$ out $\quad x = x^{\rm BG}(q,\hat{\tau}_c) + u_x^{\rm BG}(q,\hat{\tau}_c)(\tau - \hat{\tau}_c) + \delta x^{\rm out/in}$

$\tau > \tau_c(q)$ in $\quad u = u^{\rm BG}(q,\hat{\tau}_c) + \delta u^{\rm out/in}$



➡ $\delta u^{\rm out/in}(Q,\tau) = \int_{\rm out/in} d\tau'\ \delta F_{\rm ms}^{\rm out/in}, \quad \delta x(Q,\tau) = \int d\tau'\ \delta u$

➡ $\delta F_{\rm ms}(Q,\tau)$: back reaction from multi-stream (mass ordering change)

$q_1 < q_2 < q_3$ Before $\rightarrow\ x_1 < x_2 < x_3$

After $\rightarrow\ x_2 < x_1 < x_3$

➡ $\nabla_x^2 \Phi = \frac{3}{2}\Omega_{\rm m0} H_0^2\, a\, \delta(x)$ (Green function)

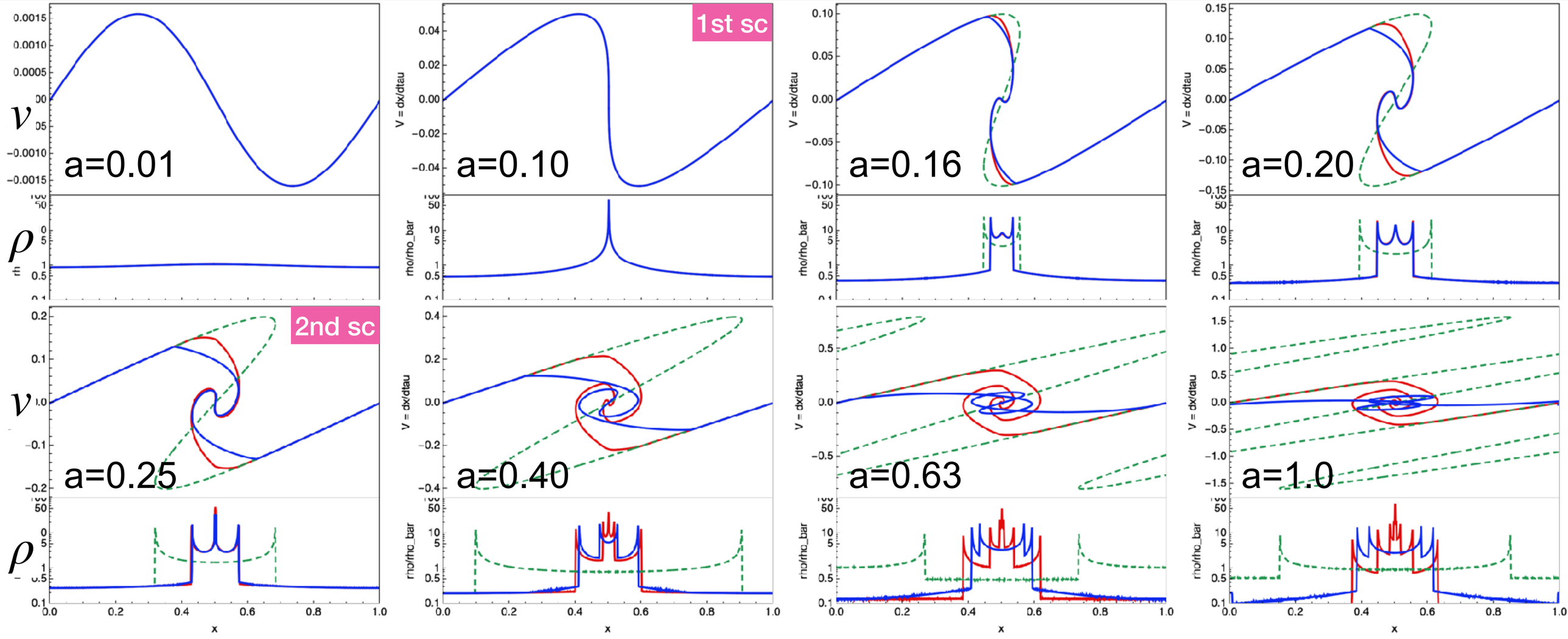
Ballistic approximation: shortly after shell crossing, keep only linear time evolution

Compare with simulation: basic PCPT in 1D

Einstein-de Sitter Universe, $\delta_{\rm L} \propto \cos(kx)$,

1st: $a \simeq 0.1$, 2nd $a \approx 0.25$

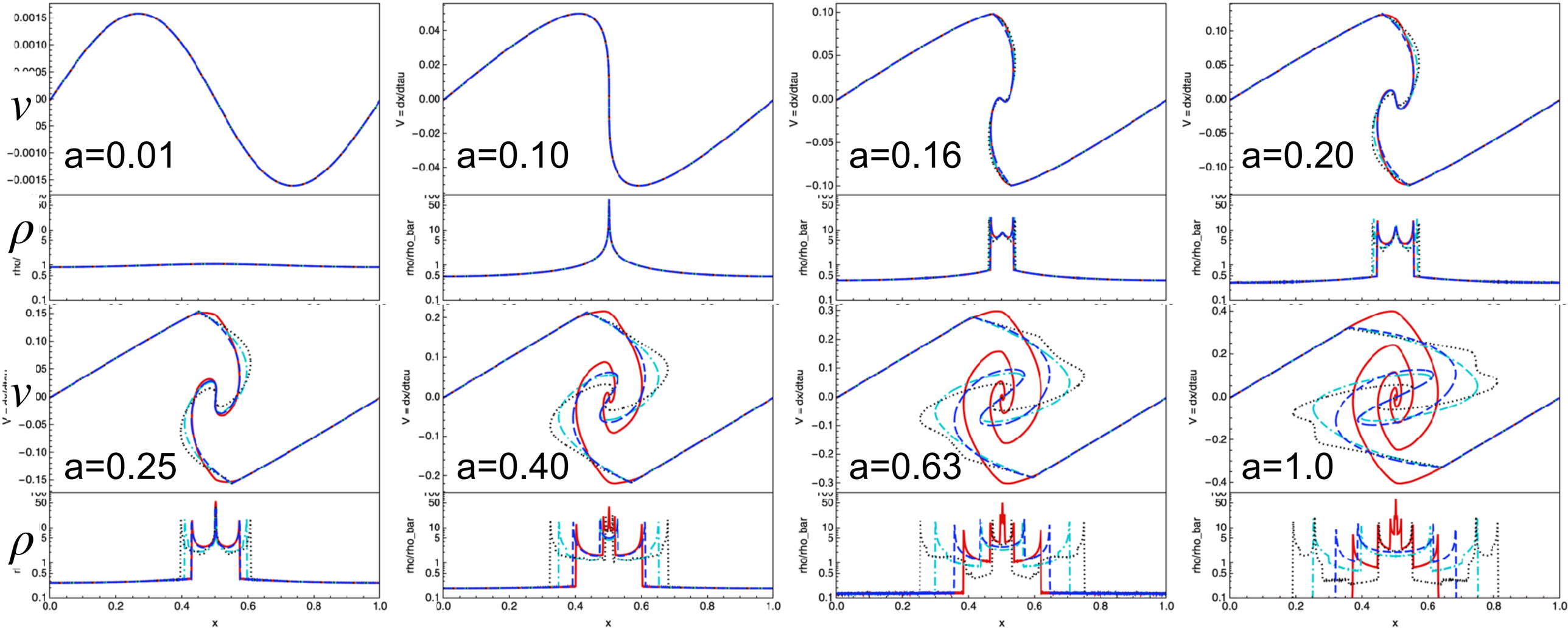
Simulation PCPT Zel'dovich



Compare with simulation: improved 1D PCPT

- higher-order critical times/boundaries $(Q_c, \hat{Q}_c, \tau_c, \hat{\tau}_c)$
- Inner PCPT → cubic spline matching → Zel'dovich

Simulation PCPT



Summary & 1D → 3D

LPT background x^{BG}

single stream

↓

Local expansion near first crossing

$J = 0$

↓

Multi-stream geometry / caustic

↓

$F_{\mathrm{multi-stream}},\ \delta u,\ \delta x$

PCPT correction

$$x^{\mathrm{BG}} - A \simeq -BQ + CQ^3$$

cubic in space

$$x = x^{\mathrm{BG}}(q, \hat{\tau}_c) + u_x^{\mathrm{BG}}(q, \hat{\tau}_c)(\tau - \hat{\tau}_c) + \delta x^{\mathrm{out/in}}$$

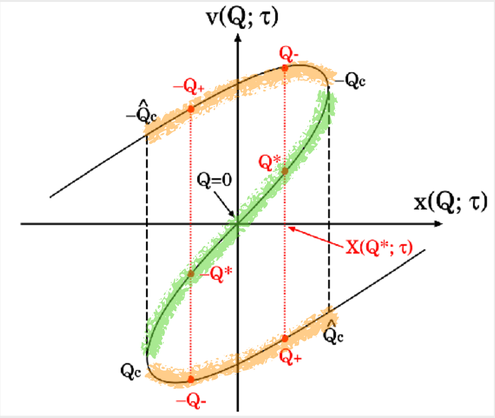
ballistic / linear in time

$$x^{\mathrm{BG}} = -B_{\mathrm{eff}} q_x + C_x q_x^3$$

BG: Zel'dovich

→ 15LPT $\boldsymbol{\psi}(\boldsymbol{q}, \tau) = \sum_{n=1}^{\infty} \boldsymbol{\psi}^{(n)}(\boldsymbol{q}, \tau)$

$\tau < \hat{\tau}_c(q)$ | $\hat{\tau}_c < \tau < \tau_c$ | $\tau > \tau_c(q)$



3D volume → pancake → filament → node/halo

thin in x + extended in y, z

Pancake collapse ($\lambda_x \to 0,\ \lambda_y \neq 0,\ \lambda_z \neq 0$)

- Local background flow near first shell crossing, collapse is locally quasi-1D

- Assume $x\,(q_x, q_y, q_z) = -\,x\,(-q_x, q_y, q_z) = x\,(q_x, -\,q_y, q_z)$

  around $\boldsymbol{q}_{\rm sc} = 0$,



Linear collapse

Transverse modulation

Caustic formation

$$x^{\rm BG} = -\,B_x q_x + (C_y q_y^2 + C_z q_z^2) q_x + C_x q_x^3, \quad y^{\rm BG} = B_y q_y, \quad z^{\rm BG} = B_z q_z$$

- Fixed $q_y, q_z \to\ x^{\rm BG} = -\,B_{\rm eff} q_x + C_x q_x^3, \quad B_{\rm eff} = B_x - C_y q_y^2 - C_z q_z^2$

3D pancake locally reduces to the same cubic form as 1D PCPT

- Caustic surface: $J = \dfrac{\partial(x, y, z)}{\partial(q_x, q_y, q_z)} = B_y B_z(-B_{\rm eff} + 3C_x q_x^2) = 0$

$$\psi_i^{\rm init}(\boldsymbol{q}) \propto \epsilon_i \sin(kq_i)$$

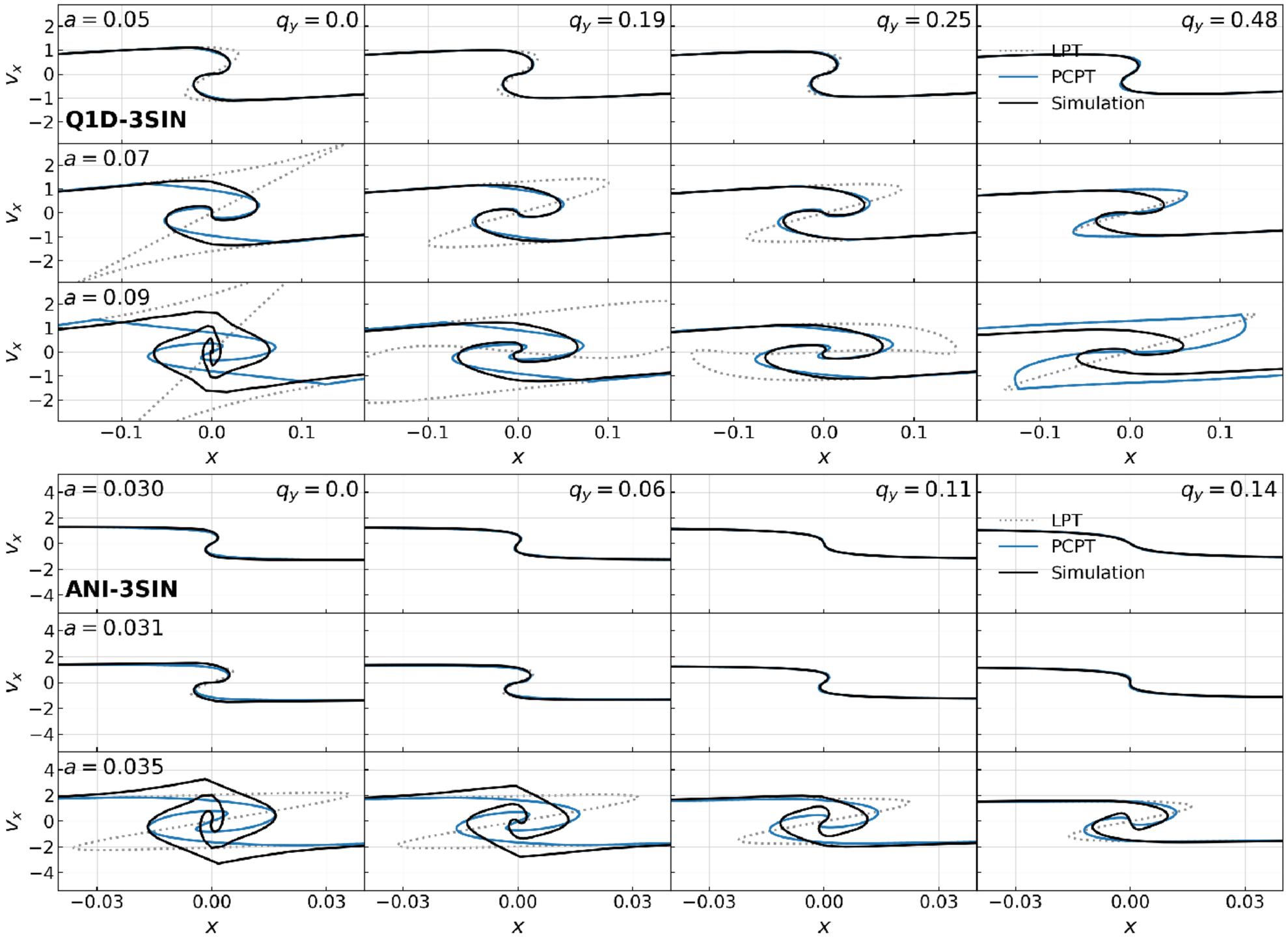
Q1D: quasi-1D

$$\frac{\epsilon_y}{\epsilon_x} = \frac{1}{6}, \frac{\epsilon_z}{\epsilon_x} = \frac{1}{8}$$

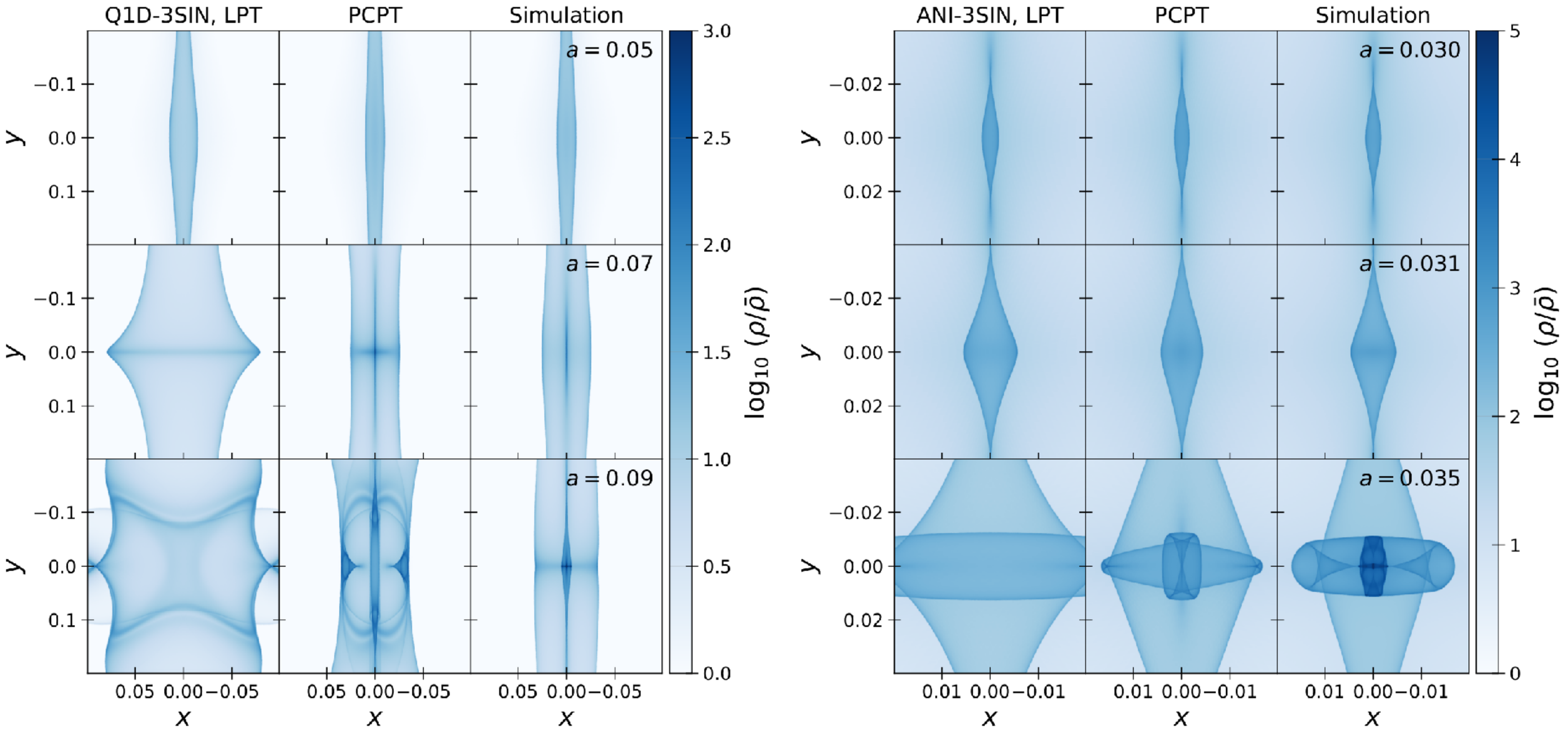
ANI: anisotropic

$$\frac{\epsilon_y}{\epsilon_x} = \frac{3}{4}, \frac{\epsilon_z}{\epsilon_x} = \frac{1}{2}$$

* $q_z = 0$
* only basic PCPT



Eulerian space density morphology



Summary / Potential improvements

What PCPT does

- Leading post-collapse correction
- Explicitly computes backreaction from multi-streaming near 1st crossing

Current 3D formulation

- High-order LPT background
- Cubic in space; ballistic in time
- Quasi-1D pancake

Improvements

- UV-completed / resummed background
- Higher-order space/time correction
- Full transverse dynamics

Thanks for listening!

1. kek 很少人cosmo

2. 一页太满，留白